

UNIVERSITY SEMESTER EXAMINATION

Subject: Artificial Intelligence & Robotics

Q5. Describe the A* search algorithm and explain its heuristic admissibility condition.

Q6. Formulate the Minimax algorithm for two-player zero-sum games with Alpha-Beta pruning.

Explain how Alpha and Beta bounds are maintained during depth-first search traversal and provide

a complete pseudocode implementation showing cut-off conditions and move evaluation.

Q7. What is the Markov Decision Process (MDP) framework? Define Bellman optimality equations.